



5.

$$\mathbf{M} = \begin{pmatrix} a & 2 & -3 \\ 2 & 3 & 0 \\ 4 & a & 2 \end{pmatrix} \quad \text{where } a \text{ is a constant}$$

(a) Show that $\mathbf{M}$ is non-singular for all values of a .

(2)

(b) Determine, in terms of a , $\mathbf{M}^{-1}$

(4)

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4. Prove by induction that for $n \in \mathbb{N}$

$$\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & -2n \\ 0 & 1 \end{pmatrix}$$

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8. A colony of small mammals is being studied.
In the study, the mammals are divided into 3 categories

N (newborns)	0 to less than 1 month old
J (juveniles)	1 to 3 months old
B (breeders)	over 3 months old

- (a) State one limitation of the model regarding the division into these categories.

(1)

A model for the population of the colony is given by the matrix equation

$$\begin{pmatrix} N_{n+1} \\ J_{n+1} \\ B_{n+1} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 2 \\ a & b & 0 \\ 0 & 0.48 & 0.96 \end{pmatrix} \begin{pmatrix} N_n \\ J_n \\ B_n \end{pmatrix}$$

where a and b are constants, and N_n , J_n and B_n are the respective numbers of the mammals in each category n months after the start of the study.

At the start of the study the colony has breeders only, with no newborns or juveniles.

According to the model, after 2 months the number of newborns is 48 and the number of juveniles is 40

- (b) (i) Determine the number of mammals in the colony at the start of the study.
(ii) Show that $a = 0.8$

(4)

- (c) Determine, in terms of b ,

$$\begin{pmatrix} 0 & 0 & 2 \\ 0.8 & b & 0 \\ 0 & 0.48 & 0.96 \end{pmatrix}^{-1}$$

(3)

Given that the model predicts approximately 1015 mammals **in total** at the start of a particular month, and approximately 596 **newborns**, 464 **juveniles** and 437 **breeders** at the start of the next month,

- (d) determine the value of b , giving your answer to 2 decimal places.

(3)

It is decided to monitor the number of **newborn** males and females as a part of the study. Assuming that 42% of newborns are male,

- (e) refine the matrix equation for the model to reflect this information, giving a reason for your answer.
(There is no need to estimate any unknown values for the refined model, but any known values should be made clear.)

(2)

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8.

$$\mathbf{A} = \begin{pmatrix} 3 & 1 & -1 \\ 1 & 1 & 1 \\ k & 3 & 6 \end{pmatrix} \quad k \neq 0$$

(a) Find, in terms of k , $\mathbf{A}^{-1}$ (4)

(b) Determine, in simplest form in terms of k , the coordinates of the point where the following planes intersect.

$$3x + y - z = 3$$

$$x + y + z = 1$$

$$kx + 3y + 6z = 6$$

(3)



7.

$$\mathbf{M} = \begin{pmatrix} 2 & -1 & 1 \\ 3 & k & 4 \\ 3 & 2 & -1 \end{pmatrix} \quad \text{where } k \text{ is a constant}$$

- (a) Find the values of k for which the matrix $\mathbf{M}$ has an inverse.

(2)

- (b) Find, in terms of p , the coordinates of the point where the following planes intersect

$$2x - y + z = p$$

$$3x - 6y + 4z = 1$$

$$3x + 2y - z = 0$$

(5)

- (c) (i) Find the value of q for which the set of simultaneous equations

$$2x - y + z = 1$$

$$3x - 5y + 4z = q$$

$$3x + 2y - z = 0$$

can be solved.

- (ii) For this value of q , interpret the solution of the set of simultaneous equations geometrically.

(4)

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6.

$$\mathbf{M} = \begin{pmatrix} k & 5 & 7 \\ 1 & 1 & 1 \\ 2 & 1 & -1 \end{pmatrix} \quad \text{where } k \text{ is a constant}$$

(a) Given that $k \neq 4$, find, in terms of k , the inverse of the matrix $\mathbf{M}$.

(4)

(b) Find, in terms of p , the coordinates of the point where the following planes intersect.

$$2x + 5y + 7z = 1$$

$$x + y + z = p$$

$$2x + y - z = 2$$

(3)

(c) (i) Find the value of q for which the following planes intersect in a straight line.

$$4x + 5y + 7z = 1$$

$$x + y + z = q$$

$$2x + y - z = 2$$

(ii) For this value of q , determine a vector equation for the line of intersection.

(7)

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2.

$$\mathbf{A} = \begin{pmatrix} 4 & -2 \\ 5 & 3 \end{pmatrix}$$

The matrix $\mathbf{A}$ represents the linear transformation M .

Prove that, for the linear transformation M , there are no invariant lines.

(5)

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3.

$$\mathbf{M} = \begin{pmatrix} -2 & 5 \\ 6 & k \end{pmatrix}$$

where k is a constant.

Given that

$$\mathbf{M}^2 + 11\mathbf{M} = a\mathbf{I}$$

where a is a constant and $\mathbf{I}$ is the 2×2 identity matrix,

(a) (i) determine the value of a

(ii) show that $k = -9$

(3)

(b) Determine the equations of the invariant lines of the transformation represented by $\mathbf{M}$.

(6)

(c) State which, if any, of the lines identified in (b) consist of fixed points, giving a reason for your answer.

(1)

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